

DM879 Artificial Intelligence

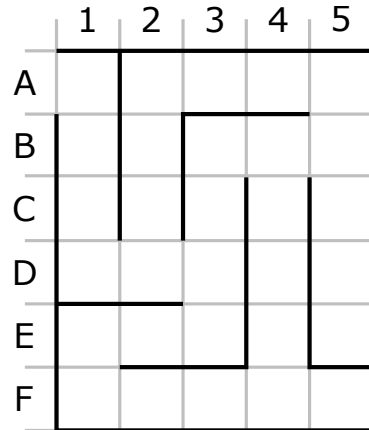
Reexam — August 27th, 2021 — Duration: 3 hours

You do not need to justify your answers unless explicitly asked to

1 The labyrinth, part I

The figure on the right depicts a labyrinth. The black lines are the walls, and for convenience a light gray grid was added in order to make it easy to refer to cells. The goal of this problem is to find a path from A1 to F5.

One possibility is for example to take the route A1-B1-C1-D1-D2-C2-B2-A2-A3-A4-A5-B5-B4-C4-D4-E4-F4-F5, which we can abbreviate to A1-D1-D2-A2-A5-C5-C4-F4-F5 (by indicating only the places where we change direction), but of course other, shorter, paths are available.



- Classify this problem according to the following dimensions: fully/partially observable; deterministic/stochastic; discrete/continuous.
- Formulate this problem precisely as a search problem: describe the state space, the initial state, the actions, the transition model, the goal test, and the cost function.
- Assume that the possible moves are generated in the order up, left, down, right, and that duplicate states are skipped.
 - Write the solution found by depth-first search.
 - Write the solution found by breadth-first search.
 - Write the solution found by bidirectional search using uniform-cost search.
 - Propose a (simple) admissible heuristic for this problem.
 - Write the solution found by best-first search using your heuristic.
- Formulate this problem as a planning problem in PDDL style: introduce the relevant predicates for describing the state, and indicate the preconditions and effects of action up (moving one step in the “upwards” direction).

2 The student’s revenge

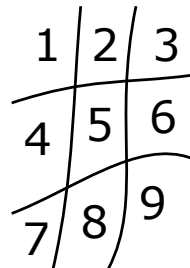
After a series of defeats playing tic-tac-toe against the teachers, one diligent student decided to implement an agent for playing the game. As a first step, the student focused on representing the game using first-order logic. The representation is as follows.

- The squares are represented by numbers (1–9), as in the diagram on Figure (a) below.
- There is a (unary) predicate Empty such that $\text{Empty}(N)$ holds iff square N is empty.
- There is a (binary) predicate Marked such that $\text{Marked}(P, N)$ holds iff square N is marked with the symbol P .

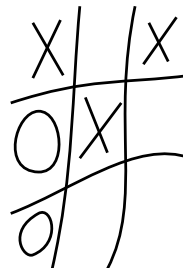
- There are auxiliary (unary) predicates Turn and Won such that $\text{Turn}(P)$ holds iff the next player is the player playing symbol P , and $\text{Won}(P)$ holds iff the player playing symbol P has won.

In all the examples above, the domain for variable P is the set $\{X, O\}$.

Important. This exercise assumes the international (a.k.a. “normal”) way of playing tic-tac-toe, where players alternate playing 'X' and 'O' on an empty square until one of them reaches three-in-a-row (win) or all squares are filled (draw).



(a)



(b)

- Give the complete set of facts that describe the initial state of a new game.
- Give the complete set of facts that describe the state in Figure (b) above.
- Write two formulas stating that (i) a player wins by marking all cells in the first row and (ii) a player wins by marking all cells in the main diagonal.
- All possible actions in this game can be written as $\text{Play}(P, N)$, corresponding to player with symbol P playing in square N . Write the PDDL specification of this action in the language used in this exercise.
- Write the four possible moves for the next player in the game depicted in Figure (b) above and show that for each of them there is a rational response leading to a win by the player playing X. (Your argument should use the effects of the relevant actions to describe the resulting states, showing that $\text{Win}(X)$ holds in all of them.)

3 The labyrinth, part II

Now we consider the scenario where the agent does not know anything about the labyrinth except for its size, and has to find a path from cell A1 to cell F5.

Model this problem as a Reinforcement Learning process specifying:

- the task environment with respect to the dimensions: fully/partial observable, deterministic/stochastic, episodic/sequential, discrete/continuous;
- the elements defining the RL process for this setting;
- an example of solution to the problem (a wordy description is also acceptable provided it is precise enough to avoid ambiguities).

Provide also a qualitative description on how the agent will find a solution to the problem.